

Novel sustained-release implant of herb extract using chitosan

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Abstract

Chitosan (CS), a naturally occurring high-molecule polymer, which is stable in vivo, has proved to be a useful biomaterial. The extract of danshen (*Radix Salviae miltiorrhizae*), a medicinal herbal, was developed with CS–gelatin as an implant for the promotion of anastomosing and healing on muscles and tissues at the organic incision site in abdominal cavities.

Measurements were made of the sustained release of tanshinone IIa, a marker component, from the material in vitro. The dissolution medium was assayed with an high-performance liquid chromatography method. Biodegradation studies of the material were also conducted both in vitro and in vivo. The film made of this material exhibited a sustained release effect. The release profile confirms to the Higuchi equation. At most about 20% of the incorporated drug were released over 15 days in a CS–gelatin (1:2) matrix. Drug release was found to be effectively controlled by the drug-amount loaded in the matrix. The improved film (CS/gelatin ratio: 1:16) can be hydrolyzed by lysozymes in vitro in 4 days. This film of 0.5 cm² was implanted and degraded completely in rats over 28 days and the animals' wounds of abdominal incision healed well. © 2002 Elsevier Science Ltd. All rights reserved.

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1. Introduction

Chitosan (CS) [$\alpha(1-4)$ 2-amino-2-deoxy-beta-D-Glucan] is a unique polysaccharide derived from chitin. Several attempts have been made to use this biopolymer in biomedical field [1]. Recently, they have attracted much interest in the biomedical industry because of their biodegradability, biocompatibility and accelerated wound-healing properties [2,3]. Furthermore, CS has been used for the production of controlled release implant systems [4,5].

This study was an attempt to develop implantable sustained release dosage forms of a drug based on matrices containing CS for the promotion of anastomosing and healing on muscles and tissues at the organic incision site in abdominal cavities.

Danshen (*Radix Salviae miltiorrhizae*) is a herbal medicine and can promote blood circulation to remove blood stasis in traditional Chinese medicine. Tanshinone IIa is a lipophilic diterpene compound found in danshen as a marker component. The drug was applied in surgical clinical practice for wound healing [6] and we try to use it in the newly formulated CS implant.

2. Material and methods

2.1. Chemicals

CSs with molecular weight > 600,000 and degree of deacetylation > 90% were obtained from Shuanglin Bio-products Co. Ltd. (Nantong, China). Danshen was purchased from the local pharmacy, and identified as radix of *Salviae miltiorrhizae* Bge (*Labiatae*) by Wang. Tanshinone IIa was used as reference, supplied by Central Control Institute of Pharmaceutical and Biological Products (Beijing, China). Lysozyme was the biochemical reagent (Boao Biologic Science & Technology Co., Shanghai, China). Gelatin, acetic acid, ethanol, methanol, PEG 400 and chloral hydrate were purchased from common commercial sources, among which methanol was of high-performance liquid chromatography (HPLC) grade and other reagents were analytically pure.

2.2. Apparatus

The HPLC was carried out using a 10A HPLC system (Shimadzu, Japan) with a LichroCART RT-C₁₈ column (250 × 4.6 mm; 5 μ m) (Merk, Germany). A dissolution

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test instrument (ZRS-4, Radio Factory, Tianjin University, China) was used in the *in vitro* release and biodegradation studies.

2.3. Animals

Wistar rats (Animal Center, China Pharmaceutical University).

2.4. Extract of danshen

Air-dried and powdered danshen roots (250 g) were soaked in 1500 ml of 95% ethanol for 60 min, and then refluxed twice in a water bath for about 2 h. After filtration with paper the extracting liquor was combined and then filtrated through an ultra-membrane filter. The filtrate was concentrated *in vacuo* to obtain the danshen extract for the preparation of the film.

2.5. Preparation of films

CS solution (2%) was prepared by dissolving CS powder in 1.5% acetic acid with stirring. The CS solution was mixed well with 8% gelatin solution. After being filtrated with paper and degassed with ultrasonic, the mixture was poured into a mould (12 × 12 cm). After cooling and drying, the matrix was formed with a thickness of about 1 mm. The preparation of danshen film was carried out with the same method for shaping as above. The CS solution/gelatin solution ratio was improved gradually in the film for the following tests.

2.6. *In vitro* release study

The *in vitro* release study was conducted using saline containing 20% ethanol and 30% PEG 400 (pH 5.0) as dissolution medium as follows [7], since tanshinone IIa was difficult to dissolve in water. The danshen films (12 × 12 cm², CS/gelatin ratio: 1:2) at different doses (5.0, 1.0, 1.5 mg of tanshinone IIa) were placed in a glass beaker with a cap containing about 100 ml of the dissolution medium, and the beaker was incubated at 37.5°C with a roller rotating at 50 rpm. At predetermined time, the dissolution medium was withdrawn and the fresh medium was added. Each experiment consisted of 3–4 samples.

2.7. Determination of tanshinone IIa by HPLC

The conditions were applied according to the method for assay of danshen in Chinese Pharmacopoeia (2000 edition). The mobile phase was methanol–water (75:25). The analyses were carried out at a flow rate of 1.0 ml/min with UV detection at 270 nm for tanshinone IIa in each of the dissolution medium.

2.8. *In vivo* biodegradation study

Three female Wistar rats weighting 260–290 g were used to observe biodegradation of the matrix (CS/gelatin ratio: 1:6) for 1 week before a large test of biodegradation in rats. The rat was anesthetized with an ip injection of chloral hydrate (300 mg/kg). The rat was shaved at the surgical site, and an incision of 1.5 cm long was operated at the midline on the belly. A matrix implant (0.5 × 1.0 cm) was implanted in the abdominal cavity by suturing. Then the wound was closed in standard surgical fashion.

A total of 16 animals divided into four groups of 4 rats each was used to test biodegradability for the matrix (CS/gelatin ratio: 1:16) in four times (7, 14, 21, and 28 days). The rat was anesthetized with an ip injection of chloral hydrate (300 mg/kg). The rat was shaved at the surgical site, and an incision of 1.5 cm long was operated at the midline on the belly. A film implant (0.5 × 1 cm) was implanted in the abdominal cavity by suturing on the front stomach wall. Then the wound was closed in standard surgical fashion. After the operation, the rat was reared in-house with standard feedstuff, and killed at predetermined times. The film was tore carefully, then immersed, dried, and weighed. Biodegradability of the film was calculated as follows:1

$$\text{Degraded weight percent} = (W_1 - W_2)/W_1 \times 100\%,$$

where W_1 is the weight before degradation, W_2 is the weight after degradation.

2.9. *In vitro* biodegradation study

In vitro biodegradation study was conducted with hydrolysis of the danshen film by lysozyme. Four pieces of films (CS/gelatin ratio: 1:16; 5 mg of tanshinone IIa loaded in each film) were placed in a glass beaker with a cap containing 100 ml of lysozyme saline solution (1 mg/ml), and the beaker was incubated at 37.5°C with a roller rotating at 50 rpm. Normal saline and 30% PEG 400 saline was used as the reference medium for two pieces of danshen films, respectively. Degradation was then observed.

3. Results and discussion

3.1. *In vitro* release profile

The *in vitro* release profile of tanshinone IIa from the film matrix was monitored, as a function of time, in saline solution containing 20% ethanol and 30% PEG 400 (Fig. 1). A very small release rate was observed with the matrix loading tanshinone IIa (Fig. 2). The release of drug from the solid dispersions obviously varied depending on the amount of the drug loaded in the

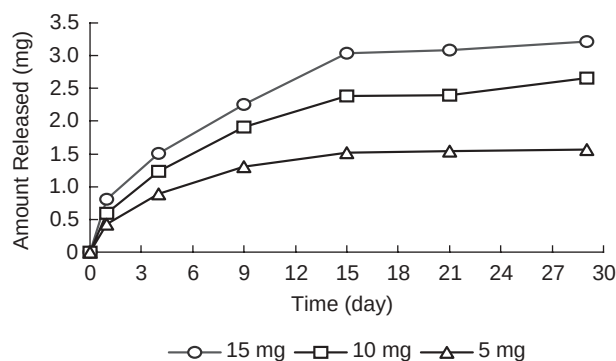


Fig. 1. Release from danshen films with different tanshinone IIA loaded in vitro.

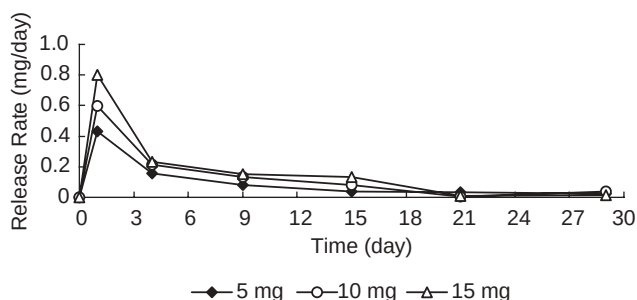


Fig. 2. Release rate from danshen films with different tanshinone IIA loaded in vitro.

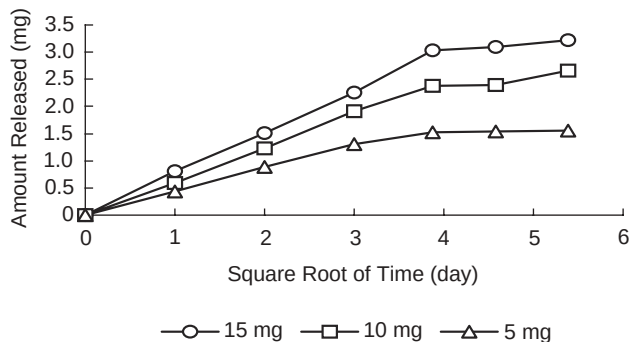


Fig. 3. Amount released from danshen films with different tanshinone IIA loaded in vitro, depending on square roots of time.

matrix. The percent released of tanshinone IIA over 15 days was 20% from the matrix loading 15 mg dose. The amount released of the active compound was linear corresponding to the square root of the time during 0–15 day, as shown in Fig. 3. The release profile confirms to the Higuchi equation:

$$\text{Amount released} \approx A(2DC_L C_S t)^{1/2}, \text{ if } C_L \gg C_S,$$

where A is the cross-sectional surface area; D is the diffusion coefficient of drug; C_L is the loading concentration of drug in the matrix; C_S is the saturation concentration of drug in the aqueous media. This equation predicts that the square root of time depends on the amount of the drug released from a polymeric

Table 1

Weight loss of drug loaded films in abdominal cavities of rats

Time (day)	Average weight of film $n = 4$ (mg)	Degradation rate (mg/day)	Weight loss (%)
0	54.70 ± 1.233	0	0
7	28.33 ± 1.520	3.77	48.2
14	14.00 ± 2.058	2.05	74.5
21	4.27 ± 0.520	1.39	92.2
28	0	0.61	100

matrix [8], and the danshen film exhibits a sustained-release effect.

3.2. Biodegradability in vivo

The CS (molecular weight > 600,000)–gelatin film was a better choice for making a matrix implanted in the abdominal cavity. The pretest of biodegradability showed that after 1 week some of the matrix in the abdominal cavity, in which the relative content of gelatin was increased, was wrapped up, and easily broken when the adhesions were opened. A total of three rats tested were in good condition and the wound on the abdomen had healed. No inflammatory response was observed.

The degradation of CS–gelatin films, in which CS/gelatin ratio changed from 1:6 to 1:16, was also investigated by weight loss in rats (Table 1). The implanted film of 0.5 cm^2 was degraded completely in 28 days. The degradation rate declined as the time was prolonged when the film was implanted in the abdominal cavity. The rats of the four groups did not present any abnormality and their wounds of abdominal incision healed well after 7 days of surgery. Adhesions between intestinal tubes had been absorbed, but the peritoneal adhesions still occurred on the 14th, 21st and 28th days after operation.

3.3. Biodegradation in vitro

The improved film (CS/gelatin ratio: 1:16) can be hydrolyzed by lysozymes. The film pieces immersed in the lysozyme saline solution (1 mg/ml) were gradually bulged, broken, and dissolved in 4 days, compared with the saline reference, 50% of which destroyed in the normal saline over 12 days, but the films in 30% PEG400 saline were not broken obviously.

In this present study, a sustained-release dosage form of an herbal medicine was produced using CS–gelatin as the matrix for implanting in the abdominal cavity. The release of tanshinone IIA from the matrices was initially high, but from which the percentage of drug released in 15 days was only 20% for 15 mg dose, since the relative content of CS was higher in the first

formulation and the dissolution medium used was not very available. As the content of gelatin was increased in the following formulations, the film became more soluble and the degradation of the film was accelerated both in vivo and in vitro. The biodegradability, of the CS–gelatin matrix and drug loaded CS–gelatin films, was suitable to the medical application and no side effect was observed. Because of bioabsorption of such implants, there is no need for a removal operation after healing of the tissue. However, the release profile should be further studied with a simulated body fluid. The effect of treatment will be reported, using the danshen film compared with implanted matrix without drug and wound alone.

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